

Washbasin MIRROR LIGHT CONTROLLER

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This controller circuit based on an infrared (IR) object sensor (Sharp 2Y0A21) is used for saving power of the mirror light near a washbasin. The circuit contains manual on and off switches and an additional automatic switching system.

Circuit and working

The circuit diagram of the washbasin mirror light controller is shown in Fig. 1. It is built around bridge rectifier BR1, 5V voltage regulator 7805 (IC1), LM358 op-amp (IC2), NE555 timer (IC3), two BC548 transistors (T1 and T2), 12V single-changeover relay (RL1), object sensor (Sharp 2Y0A21) and a few other components. If you forget to turn off the light, the circuit automatically turns it off after a few seconds.

Light bulb B1 connected across connector CON3 is turned on with the help of switch S1 and turned off using S2 after using the washbasin. LM358 (IC2) is configured as a comparator. Its non-inverting input (pin 3) and inverting input (pin 2) are used as inputs. Pin 2 of IC2 is connected to voltage-divider

potmeter VR1. Using VR1, vary the reference voltage at pin 2. Initially, output pin 1 of IC2 is low.

NE555 (IC3) is designed as a monostable multivibrator. On time of IC3 is adjusted to eleven seconds with the help of timing capacitor C4 and resistor R4. Output pin 3 of IC3 is fed to relay driver transistor BC548 (T2).

When S1 is pressed, IC3 is triggered and, hence, its output pin 3 goes high, and mirror light bulb turns on through relay (RL1) contacts. When S2 is pressed after use, the light turns off.

If you forget to switch the light off, the sensor comes into action. Object sensor output is connected to inverting pin 2 of IC2. Under normal conditions, voltage at pin 3 of IC2 is low. When a person comes in front of the sensor connected across CON2, voltage at pin 3 is higher than voltage at pin 2 of IC2. Hence, output pin 1 of LM358 is high. This output is connected to the base of transistor T1. Collector and emitter of T1 are connected across C4. Discharging time of C4 extends the on time of the light bulb.

PARTS LIST

Semiconductors:

IC1	- 7805, 5V voltage regulator
IC2	- LM358 op-amp
IC3	- NE555 timer
T1, T2	- BC548 npn transistor
D1	- 1N4007 rectifier diode
ZD1	- 12V, 1W Zener diode
BR1	- 1A bridge rectifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon) unless stated otherwise:

R1	- 1-mega-ohm
R2, R3	- 100-ohm, 1W
R4	- 100-kilo-ohm
R5, R8	- 1-kilo-ohm
R6, R7	- 10-kilo-ohm
VR1	- 10-kilo-ohm potmeter

Capacitors:

C1	- 0.1μF, 450V polyester
C2	- 470μF, 25V electrolytic
C3	- 47μF, 16V electrolytic
C4	- 100μF, 16V electrolytic
C5	- 0.01μF ceramic disk

Miscellaneous:

CON1	- 2-pin terminal connector
CON2	- 3-pin connector
CON3	- 2-pin connector
S1-S2	- Tactile switch
RL1	- 12V single changeover relay
B1	- 60W, 230V AC bulb
	- Sharp 2Y0A21 object sensor
	- 230V AC, 50Hz power supply

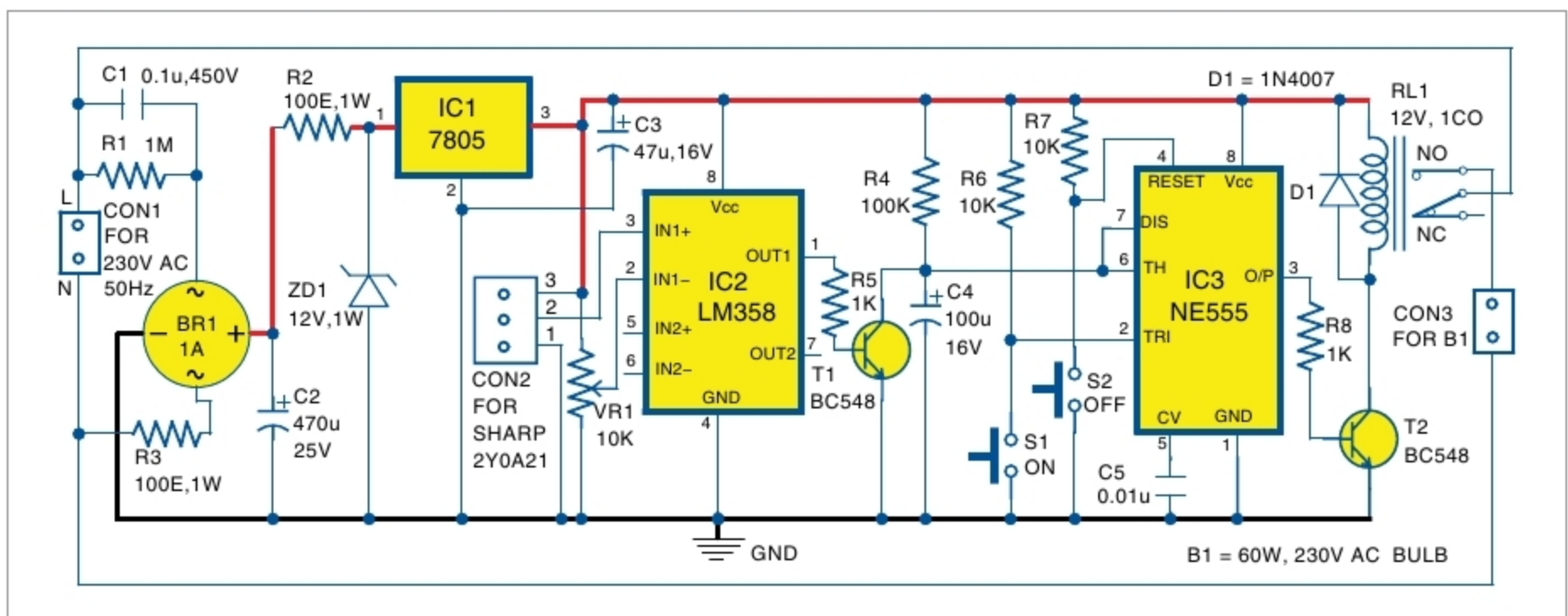


Fig. 1: Circuit diagram of washbasin mirror light controller